Ganga Singh

FULL-STACK SOFTWARE ENGINEER

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TECHNICAL SKILLS

LANGUAGES

JavaScript (ES6+) · TypeScript · C++ · Python · SQL

FRONTEND

React.js · Next.js · HTML5 · CSS3 · Tailwind CSS

BACKEND

Node.js · Express.js · REST APIs · WebSockets · Prisma ORM

DATABASES

PostgreSQL · MongoDB · Redis

DEVOPS & TOOLS

Docker · Git · GitHub · Postman · Linux · Vercel · Netlify

CONCEPTS

Data Structures · OOP · Caching · System Design · Agile · SEO

EDUCATION

B.Tech, Computer Engineering
Guru Nanak Dev University
Amritsar, Punjab | 2023 – 2027

CERTIFICATIONS

- **Meta Front-End Developer**
  Coursera
- **Google Cloud Fundamentals**
  GCP – Google
- **JS Algorithms & DS**
  freeCodeCamp
- **Docker Essentials**
  LinkedIn Learning
- **Full-Stack Web Development**
  100xDevs

COMPETITIONS & HACKATHONS

- **Smart India Hackathon 2025**
  Top 10 of 50+ competing teams
- **Cosmohacks 2025**
  Top 5 finish at GNDU
- **Walmart Sparkathon - 2025**
  Qualified till Round 2
- **Hacktoberfest 2025 Contributor**
  4 accepted pull requests

OPEN SOURCE

- **React.js OSS Libraries**
  3 merged PRs – bug fixes & docs
- **Node.js Rate Limiter Pkg**
  Published, 50+ npm downloads
- **Hacktoberfest 2025**
  Quality PRs across JS & TS repos

Professional Summary

Results-driven Full-Stack Software Engineer with hands-on expertise in the MERN stack, TypeScript, and distributed systems. Built and deployed **5+ production-grade applications** leveraging React.js, Next.js, PostgreSQL, and Redis, driving measurable gains in user engagement and system performance. Skilled in REST API design, database architecture, performance optimization, and scalable application development. Passionate about clean code, system design, and delivering high-impact solutions in agile environments.

Projects

Real-Time Collaborative Research Platform

Jan 2025 – Present

Node.js · Next.js · TypeScript · PostgreSQL · Redis · WebSocket · Prisma ORM

A high-performance collaborative platform enabling researchers to co-edit shared lab notebooks in real time with live presence tracking and version control.

- Architected real-time collaboration engine using **WebSocket & Redis pub/sub**, enabling **5+ concurrent editors** with <100 ms synchronization latency.
- Designed normalized **PostgreSQL schema via Prisma ORM**, modeling relational data across users, projects, notebooks, and version history.
- Built RESTful API layer with Express.js for authentication, project management, and document versioning, processing **500+ daily API requests**.
- Implemented Redis caching for session management & presence tracking, **reducing database query load by 35%** and response times by 29%.

AI-Powered Document Study Platform

Sep 2024 – Dec 2024

Next.js · TypeScript · MongoDB · Node.js · Gemini API · Tailwind CSS

An intelligent study platform leveraging AI to auto-summarize documents and generate adaptive quizzes, accelerating student learning workflows.

- Scaled to **232 active users** with **1,455 page views in 30 days**, achieving a **14% bounce rate** through frontend performance optimization.
- Integrated **Gemini API** with custom PDF parsers for AI-driven document summarization, **reducing average study preparation time by 40%**.
- Built responsive UI with Next.js & TypeScript implementing **SSR**, improving SEO scores by 39% and Core Web Vitals to a first-contentful paint <1.6s.
- Designed flexible **MongoDB document models** supporting uploads, annotations, and personalized study plans for **200+ concurrent users**.

Key Achievements

End-to-End Product Delivery

Independently architected, developed, and deployed **5+ full-stack applications** from concept to production, managing database design, API architecture, CI/CD, and frontend implementation.

Measurable User Growth

Drove organic user acquisition to **232 active users with 1,455 page views in 30 days**; optimized Core Web Vitals to sustain a **14% bounce rate**.

Real-Time Systems at Scale

Engineered a WebSocket + Redis pub/sub architecture achieving **sub-100ms latency** for multi-user concurrent editing sessions with zero data conflicts.